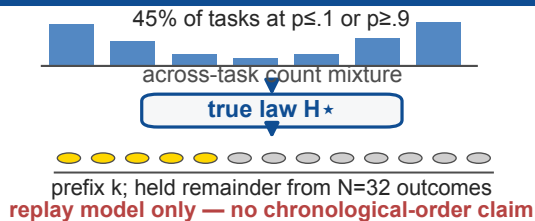


### (a) Replay prefix (random; not chronology)

random prefix • sampled without replacement



### (b) Oracle $H^*$ versus fitted score

TRUE  $H^*$  — oracle theory

$c_{H^*}(x, k) = P(\text{flip} \mid x, k)$   
exact conditional • Theorem 1

$\hat{H}_{\text{FIT}}$  — implemented fitted score

$s_{\text{HFIT}}(x, k)$ , not  $c_{H^*}$

FIT freeze → CAL EB UCB (Bonferroni)

Theorem 2: marginal replay guarantee

TEST: descriptive; no fitted conditional certificate

### (c) Replay quantities + synthetic exploratory drift

OMR  $\alpha=.05$  descriptive TEST replay-count reduction

FIXED-HOEF 15.6%

FIXED-EB 46.9%

BAYES-H 80.9% | flip 0.0249

count-replay endpoint; not correctness or operational cost

Synthetic / exploratory / nonconfirmatory drift diagnostic

E3,  $\alpha=.05$ ,  $\delta=.15$ : BAYES-H 5.2% invalid; FIXED-EB 4.7% valid.  
Explored margin  $\gamma=.025$ : flip 5.0%, reduction 77.2%; exploratory only.

Frozen JSON values; no natural-shift or online-policy claim